

Safety and ESD -- Guided Notes ANSWER KEY

Unit 1.1 | CompTIA Network+ N10-009 Obj. Core 1 5.1 | N10-008

For instructor use / distribute after students complete the notes.

Question 1

ESD stands for _____. It refers to the sudden discharge of static electricity between objects with different _____. Human hands can build up thousands of volts of static electricity -- often with no physical sensation -- yet even as little as _____ volts can permanently damage a sensitive electronic component.

Answer: Electrostatic Discharge; electrical potential (charge); 10-30 volts.

Question 2

An ESD wrist strap protects components by keeping the technician at the same _____ as the hardware being worked on. The strap contains a _____ -ohm resistor in series, which prevents the wrist strap from becoming a _____ hazard if the technician contacts live voltage.

Answer: Electrical potential (charge/ground); 1 megaohm (1,000,000); shock (electrocution).

Question 3

Antistatic bags protect components during storage and transport. A component should be placed _____ the bag, not _____ it. The outside surface of an antistatic bag can actually _____ static charge and cause ESD damage.

Answer: Inside; on top of (on the outside of); accumulate (hold / concentrate).

Question 4

Three types of equipment that contain _____ voltages and should NEVER be opened by a PC technician are: _____, _____, and _____. Even when unplugged, these devices can retain a lethal charge.

Answer: High voltages; power supply units (PSU); CRT monitors; laser printers. (Any three of these.)

Question 5

Before handling internal components, a technician should remove _____ (such as rings and bracelets) because these can bridge electrical contacts and cause _____. Working on a hard _____ floor (rather than carpet) also reduces static buildup.

Answer: Jewelry (metal jewelry); short circuits (and ESD damage); hard (non-carpeted).

Question 6

Walking across carpet can generate up to _____ volts of static electricity due to _____. When you then touch a metal object or component, the charge discharges instantly. The safest working surface for PC repair is a/an _____ mat, which dissipates charge slowly and safely.

Answer: 35,000 volts (thousands of volts); friction (triboelectric effect); antistatic (ESD).

Question 7

A latent ESD failure is one that does not cause _____ failure at the time of the discharge. Instead, the component continues to work but with _____ reliability, and may fail days, weeks, or months later. This makes latent ESD damage especially dangerous because the cause is _____.

Answer: Immediate (instant); degraded (reduced); hard to trace (difficult to identify).

Question 8 -- Real World Application

REAL WORLD: A student technician is upgrading RAM on a desktop. The workbench is carpeted. They remove the RAM from a static bag by setting the bag on the desk and placing the module on top of the bag while they prepare the system. Identify all the ESD safety mistakes in this scenario and explain how each one should be corrected.

Answer: Mistake 1 -- Carpeted surface: carpet generates static. Use an antistatic mat or work on a hard floor. Mistake 2 -- No wrist strap: technician is not grounded. Attach an ESD wrist strap to the wrist and clip to the case chassis. Mistake 3 -- Module placed on top of the static bag: the outside of the bag can accumulate charge. Store the module inside the bag. Mistake 4 -- Not touching the case before starting: should touch an unpainted metal part of the case to equalize charge before handling components.